

WORK PROCEDURE

PRO-OP-SM-003A

STOPEMASTER MOVING AND TOWING PROCEDURE

OBJECTIVE

The objective of this procedure is to define safe work methods to move and tow the STOPEMASTER

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

INTERPRETATION

The personnel having authority for the interpretation of this document is the Operations Manager. Any request for modification, revision or update must be made to this person.

RESPONSABILITIES

The supervisor is responsible for applying and following this procedure. The driller must follow this procedure and all client's procedures. The Driller is responsible for the day-to-day operation of the drill and other on-site equipment required to support the operation. The driller needs the training for all equipment he operates.

- The Driller is responsible for obtaining the maximum possible quality in any given ground condition, while achieving maximum production and maintaining a safe and efficient work environment;
- The Driller is responsible for the general condition of all site equipment and tools;
- The Driller is responsible for obtaining optimum life out of drilling tools and consumables;
- The Driller is responsible for the location of the collar of the hole, ensuring correct alignment with front and back sides, offsets, depth and setting the drill at the correct angle;

STOPEMASTER MOVING AND TOWING PROCEDURE

- The Driller is responsible for training new Drillers in safe work methods, if requested;
- The Driller is responsible for complying with all environmental regulations as legislated by local Government;
- The Driller is responsible to submit a shift report and to fill in the job's log book;

PROCEDURE

TOWING OF DRILL CARRIER WITH LHD

1. Ensure drilling area that you are leaving is free of all rubbish. Leave area safe and clean. "Leave nothing behind, except the holes";
2. Using LHD to move Stopemaster;
3. Communicate with personnel that equipment will be moved. Give locations and time;
4. Position traverse completely back and rotate to make as small as possible. Remove centralizer if required;
5. Position boom to centre of traverse;
6. Position boom angle to conform to size and shape of mine opening;
7. Make sure electrical cable is properly looped up on hangers and not dragging on ground;
8. Use proper tow bar or drill trailer/sloop;
9. Connect tow bar or drill trailer/slope to LHD;
10. Avoid positioning body between LHD and Stopemasters;
11. Ensure LHD operator has you in his line of sight;
12. Adopt a system of hand signals. Only one person can give the signals;
13. A wire rope sling or chain must be installed as a second means of protection in case of tow bar failure;

STOPEMASTER MOVING AND TOWING PROCEDURE

- 14.If the procedure of the site we operate is more severe, use the procedure of the site;
- 15.Unlock all four spring-applied brakes when using the tow bar;
- 16.Move Stopemaster towards drill location;
- 17.When following behind, keep a good distance and avoid pinch points. Visually sweep backs and walls for clearance;
- 18.Keep a watch out for vehicle traffic and/or pedestrians;
- 19.When you stop, make sure spring-applied brakes are activated before disconnecting tow bar and wire rope sling;
- 20.Never walk behind a Stopemaster while towing.

PREPARATION FOR MOVING AT DRILL SITE

1. Avoid rushing -- plan every move in advance;
2. Wear personal protective equipment;
3. Before entering drill station, refer to the JOB SITE INSPECTION;
4. Remove all obstructions or tripping hazards;
5. Layout all hoses and/or power cables. Power cables must be hung up off the ground;

WARNING!!!

Make sure drifter is at pivot point or lower before loosening hook bolt, or new assembly cone clamp!!!

6. Make sure all hoses are tightly connected to avoid air and water leaks;
7. Install whip checks on hoses 1" or greater;
- 8. Make sure all guards are in place;**
9. Blow air line free of dirt and water;
- 10.Connect air supply to air receiver;
- 11.Before charging the lines with air or water, make sure all controls are off;

STOPEMASTER MOVING AND TOWING PROCEDURE

12. Replace any frayed or worn parts;
13. Ensure ground fault system is working properly;
14. Test shut off switches;
15. Complete pre-operational checks;
16. Remove bent rods or damaged rods.
17. Position rod rack adjacent to slide out of work or travel area.

SAFE TRAMMING PROCEDURES IN THE DRILLE SITE

1. Whenever a drill is to be trammed, whether from a ring position to another or into a drill station, the area must be inspected for uneven ground, tripping hazards or things that may be sticking out of the ribs or walls. Inspect the equipment to ensure that the levers and hoses are in good working condition;
2. Inspect the air hoses to be sure they are the correct length for the move and will be out of the way. Be sure the hoses do not create a tripping hazard during the move;
3. Be sure the 600 VOLTS cable is long enough and it is not in the travel path;
4. Place the tramming levers in the correct position for the task. You must be able to see where you want to go and not position your body in the line of fire from tripping hazards or pinch points from the ribs or walls;
5. Be sure that you are clear and tramming levers are in the neutral position before powering up hydraulic pump;
6. Always tram in a controlled manner, do not try to go too fast;
7. Plan your moves before moving the drill;
8. Keep a safe distance of 8 feet from the drill at all times;
9. The body of the drill must never be close enough to touch the walls, the bench or the back when you're moving.

ALWAYS KEEP A MINIMUM DISTANCE OF 2.4M BETWEEN YOU AND THE DRILL. NEVER ATTEMPT TO WALK AND TRAM AT THE SAME TIME.

STOPEMASTER MOVING AND TOWING PROCEDURE

TRAMMING HX STOPEMASTER (DIESEL)

ALWAYS STAND CLEAR OF THE DRILL DURING DIESEL TRAMMING OF DRILL. NEVER PLACE YOURSELF BETWEEN THE DRILL AND THE WALLS DURING MOVES. NEVER ALLOW YOURSELF TO GET PINCHED BETWEEN THE DRILL CARRIER AND THE WALLS

ON THE RAMP

1. Complete preventative maintenance check on the drill;
2. Hook up the 600 VOLTS cable on the rack;
3. Turn "ON" master switch;
4. Verify that emergency stops are in the "UP" position on the tram remote and operator control panel;
5. Position the selector switch on the main electrical panel to "DIESEL";
6. Turn ignition key to start the engine while depressing oil override switch until engine starts. (Override switch is on the left of the ignition key switch);
7. Be sure that dump and mast are completely dumped back over drill and completely folded down;
8. Test emergency shut off buttons before proceeding to the ramp;
9. Check that the fuel tank has enough fuel for the move. Make sure that there is a set of wheel chocks on board;
10. Position the mast as you can attach the control to it;
11. Fix the control on the centralizer;
12. Wrap console hoses and console electrical cable and attach to console. Ensure they are clear of any potential damage when moving or positioning the machine;
13. Dump the feed all the way to the horizontal position on the traverse frame;
14. Attach two additional chains or slings to the bottom of the remote drill control and to each side of the machine for a secure three-point attachment;

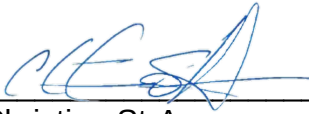
STOPEMASTER MOVING AND TOWING PROCEDURE

15. Perform Brake Test

1. Place selector switch into TRAM mode,
2. Drill can be operated in either electric or diesel mode,
3. Position yourself on the tram platform,
4. Push the BRAKE TEST BUTTON and hold it,
5. Engage the tram levers in the forward direction for 5 seconds,
6. Engage the tram levers in the reverse direction for 5 seconds,
7. Release the BRAKE TEST BUTTON,
8. If the unit moved during the test, tag it out and call the supervisor.

16. Before entering the ramp, be sure you have the priority and there is no traffic from the other way;
17. Advance slowly and cautiously to the ramp;
18. **The tramming platform must always be up ramp;**
19. Proceed very slowly;
20. Watch for low pipes and obstructions along the way;
21. Always use the three points of support to step down the tram platform;
22. Park and shut down the drill in a safe area;

**PARKING MUST BE DONE WITH THE STABILIZERS SUPPORTED ON THE GROUND SO THAT THE WHEELS NO LONGER TOUCH THE FLOOR.
IF IT IS IMPOSSIBLE TO PUT THE STABILIZERS, PUT THE WHEEL CHOCKS**



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